The issue of pricing of infrastructure services becomes critical here. In this sphere the long track record of uneconomic pricing and prevalence of subsidies will be major obstacles.

Demand Orientation of Services: The existing procedure of financing infrastructure facilities is based on plan allocation and is mainly supply-oriented. Insufficient stress on the existing infrastructure and the anticipated demand has resulted in deviations in different counties and consequently a large part of such investments are not providing sufficient returns. Privatisation will necessitate a demand-oriented approach.

The challenge for policy is to find appropriate market signals which indicate the future trend of infrastructure demand and to coordinate the supply of such facilities in such a manner that investment in infrastructure provides appropriate returns.

Allocation of Risk: Allocation of risk is of key importance in commercialisation of infrastructure. The risk should be appropriately demarcated and allocated to different stake-holders. This is important for two reasons:

i) There is a tendency among the private shareholders to shift the risk to the government.

ii) There is also a tendency among the shareholders to shift the risk on each other.

Direct Participation by the Government: While the existence of elements of monopoly in infrastructure will necessitate regulation by the government, constraints in financing and user charging will render the direct participation by the government necessary. Therefore, a transparent framework for promotion of synergistic firmness of public-private partnership in infrastructure is required.

A Suitable Regulatory Framework: Most of the infrastructure projects are right candidates for developing as a 'natural monopoly'. These cost per unit scale of output. But these units cannot be allowed, in the interest of public welfare, to eschew competition. A healthy guided competition is required for sound growth. This presupposes the existence of an effective regulatory system.

5.3 PHYSICAL INFRASTRUCTURE: GROWTH AND POLICY ISSUES

5.3.1 Transport Sector

Transport is an essential economic infrastructure for the rapid development of any region. The lack of transport facilities retards the process of economic development even if a region is endowed with rich natural resources. Transport has been recognised as an indispensable ingredient of a country's overall development.

5.3.1.1 Roads

India has the one of largest road network across the world, spanning over a total of 59 lakh km of road length. This road length includes National Highways (NHs), Expressways, State Highways (SHs), district roads, PWD roads, and project roads. In India, road infrastructure is used to transport over 60 per cent of total goods and 85 per cent of total passenger traffic. Road transportation has gradually increased over the years with the improvement in connectivity between cities, towns, and villages in the country.

Market size

As of March 01, 2019, the total length of National Highways in India stood at 132,499 km. Huge investments have been made in the sector with total investment increasing more than three times from Rs. 51,914 crore in 2014-15 to Rs. 158,839 crore in 2018-19. The total national highways length increased to 122,434 kms in FY18 from 92,851 kms in FY14.

However, the roads sector has been facing several issues including:

i) lack of equity with developers,

ii) higher cost of financing,

iii) shortfall in funds for maintenance,

iv) unavailability of land for the expansion of NHs,

v) significant increase in land acquisition cost,

vi) bottlenecks and checkpoints on NHs which could adversely impact benefits of GST, and

vii) the value of NPAs in the infrastructure sector (including roads and highways) has been increasing, with NPAs at around Rs. 2.6 lakh crore as of August 2016.

Issues with financing

As observed by the Standing Committee on Transport (2016), the Ministry of Road Transport and Highways does not have its own source of revenue other than budgetary support from the central government, also huge budget allocations by the central government would be unsustainable in the long-run. Hence, there is a need to set up Ministry's own dedicated financial institutions to generate funds for development of the road sector.

As noted by the Standing Committee on Transport (2018), road development needs concerted efforts in the form of mobilisation of funds from other sources as private sector involvement has been depleting in recent years.

Targets vs Performance

Achievement of construction targets (for NHs) has ranged between 55 per cent and 70 per cent during the period 2014-15 to 2019-20. The targets could not be met due to shortage of funds as noted by the Standing Committee on Transport (2017). Other reasons for incomplete projects include delays in obtaining clearances, poor financial and technical performance of the contractors, and law and order issues. The Economic Survey 2018-19 also highlighted issues, such as time and cost overruns, due to delays in project implementation, procedural delays, and lesser traffic growth than expected, which increased the risk factor of the projects resulting in stalling of projects.

Project Delays and increase in Project Costs

The Committee on Public Undertakings (2017) had noted that from 1995, till June 2016, out of the total 388 projects completed, only 55 projects were completed on or before time. Delays in the completion of the projects were mainly attributed to: (i) the long time taken in land acquisition, and obtaining environment and forest clearances, (ii) poor performance of concessionaires due to economic slowdown, (iii) cash flow problems, and (iv) law and order issues. Such delays increase project costs, eventually making certain projects unviable.

Government Initiatives

Some initiatives undertaken by the government are as follows:

- The Government of India has allocated Rs. 111 lakh crore (US$ 13.14 billion) under the National Infrastructure Pipeline for FY 2019-25. The Roads sector is expected to account for 18 per cent capital expenditure over FY 2019-25.
- Government of India approved the launch of Phase-III of its rural road programme Pradhan Mantri Gram Sadak Yojana (PMGSY) which aims at consolidation of 1,25,000 kms through Routes and Major Rural Links that connect habitations to Gramin Agricultural Markets (GrAMs), Higher Secondary Schools and Hospitals with an estimated cost of Rs. 80,250 crores for the period 2019-20 to 2024-25.
- A total of 65,000 km of roads and highways are to be constructed under Bharatmala Pariyojana.
- 100 per cent FDI is allowed under the Automatic route subject to applicable laws and regulations.
- ***Connectivity in Remote Areas:*** The Ministry of Road Transport and Highways also allocates funds towards the development of highways in areas with poor connectivity. Some of these projects include the Special Accelerated Road Development Programme in North East (SARDP-NE), Externally Aided Projects and Roads Projects in Left-Wing Extremism Affected Areas.
- In order to accelerate the pace of construction, large number of initiatives has been taken to revive the stalled projects and expedite completion of new projects by the Ministry of Road Transport and Highways. These include:

  i) Streamlining of land acquisition and acquisition of major portion of land prior to invitation of bids.

  ii) Award of projects after adequate project preparation in terms of land acquisition, clearances, etc.

  iii) Close coordination with other Ministries and State Governments

  iv) One time fund infusion

  v) Regular review at various levels and identification/removal of bottlenecks in project execution

  vi) Proposed exit for Equity Investors

  vii) Securitisation of road sector loans

  viii) Disputes Resolution mechanism revamped to avoid delays in completion of projects.